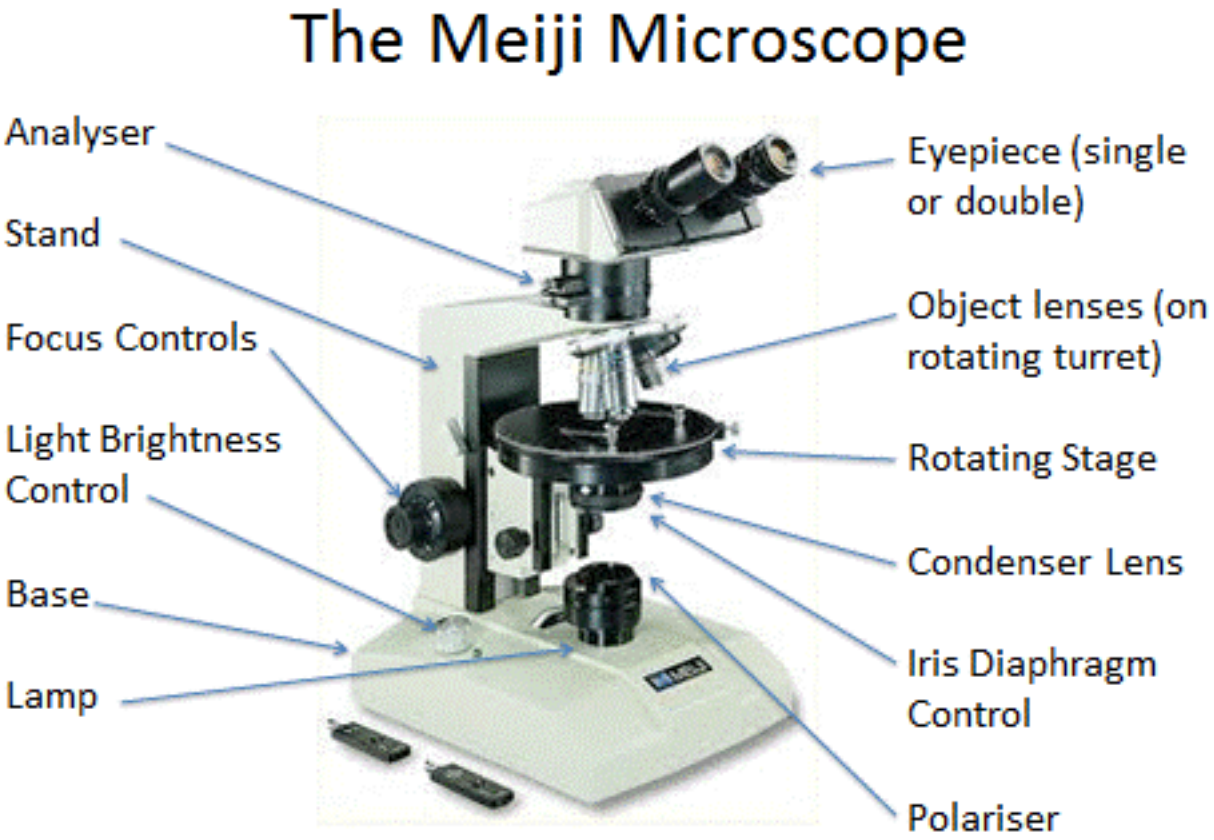
EART11300 Practical Skills: Introduction to the polarising microscope and optical properties of minerals in thin section

Practical aims:
In this practical you will:

- learn how to use the petrological microscope safely,
- be introduced to some of the optical properties of minerals in plane-polarised light – surface relief, colour, pleochroism, cleavage – and under crossed polars – isotropic and anisotropic behaviour and interference colours,
- practice drawing sketches of some of the features of thin sections under the microscope.



You will use:

- The analyser open = plane polarised light (PPL)
  closed = crossed polars (XPL)
- The focus control, to sharpen the image.
- The rotating stage, to view the sample at different angles.
- The substage iris diaphragm, to control the amount of light passing through the thin section and change the contrast.

OPTICAL PROPERTIES OF MINERALS IN THIN SECTION

Introduction

Minerals have a regular internal atomic arrangement that affects many of their properties, including their optical properties. Some of these properties vary with direction (because the internal crystal structure varies with direction), while other properties are apparently unaffected by orientation. In today's practical you will be introduced to these properties, and next week you will learn how to identify a few common rock-forming minerals on the basis of their optical properties.

1. PROPERTIES IN PLANE-POLARISED LIGHT (PP) Polariser, but not analyser in position.

a) **SURFACE RELIEF.** This property is a function of the difference between the refractive index (R.I.) of the mineral and that of the epoxy (R.I. = 1.54) which covers the surface of the thin section and acts as a cement for the cover glass. The surface appearance of minerals with different R.I. should be distinctive. Minerals with **low relief** (R.I. close to 1.54) show a rather smooth, unobtrusive surface, while those with high relief (R.I. in the range 1.6 - 1.8) display a rougher, rather frosted or pitted surface. Subtle differences in relief may be enhanced by closing the substage iris diaphragm.

(Although the R.I. of most minerals is strictly dependent on the direction taken by the light rays passing through the mineral, this is rarely detectable in terms of surface relief, which can therefore be regarded as constant for a given mineral; calcite is the only common exception to this).

b) **COLOUR.** Although most minerals are translucent in thin section, a few are still able to absorb or reflect all the light reaching them; these appear black and are referred to as **opaque.** Of the translucent minerals, many are colourless, but certain minerals preferentially absorb light of particular wavelengths and thus appear coloured as the remainder of the spectrum reaches the eye. Many of the more strongly coloured minerals (in thin section) also exhibit **pleochroism, which** is the variation in colour (intensity and/or colour tone) with respect to the orientation of the mineral. In other words, the mineral has different absorption characteristics in different directions. Pleochroism is therefore seen as colour differences from grain to grain of the same mineral and as colour changes of individual grains when the microscope stage is rotated.

The combination of relief and colour properties is usually sufficient to differentiate between the various mineral components of a rock in thin section, although further tests may be necessary before a precise identification is achieved.

c) **CLEAVAGE.** Many minerals tend to break smoothly along particular planes. This is related to the internal atomic structure: it is a manifestation of weaker bond strength across these planes. Cleavage planes are related to the crystal's internal symmetry, and are useful as a diagnostic property, since they are not affected by the external conditions that determine crystal shape. Well-developed cleavage directions can usually be recognised in thin section as evenly spaced, parallel or sub-parallel cracks. Minerals with two or more cleavages generally display evidence of some cleavage in most grains, but only in a few favourably oriented grains will the complete cleavage pattern be clear. Minerals with one cleavage will only display this in sections cut approximately perpendicular to this cleavage. Less regular cracks or **fractures** may occur quite independently of cleavage. Cleavage and fracture patterns can be accentuated by closing the substage iris diaphragm.

2. PROPERTIES UNDER CROSSED POLARS (CP or XP) Polariser and Analyser in position.

a) **ISOTROPY and ANISOTROPY.** Non-crystalline substances such as glass, epoxy, etc., together with a few minerals, are optically isotropic. This means that they appear black under crossed polars and remain black when the microscope stage is rotated. Note that most thin sections contain occasional holes that are either original features of the rock or are developed during the slide preparation. These will appear isotropic because they are filled with epoxy, but can be distinguished from genuine minerals because they have zero relief. Note also that it is impossible to tell whether opaque minerals (in thin section) are isotropic or anisotropic, since they already appear black.

The majority of minerals are anisotropic and in general these display a coloured appearance under crossed polars (but note that these colours will be 'blacked-out' at 90° intervals during rotation of the microscope stage; this phenomenon is termed **extinction**). These are termed **interference colours** and they are developed when light enters an anisotropic mineral because it splits into two rays which take slightly different paths through the crystal and get 'out-of-phase' with one another, resulting in optical interference after passing through the analyser. This splitting is termed **double refraction.** However all anisotropic minerals contain either one (uniaxial) or two (biaxial) directions of isotropic behaviour, termed **optic axes.** When light travels through the crystals in these directions, it does not undergo double refraction and therefore cannot produce interference colours. Crystal sections cut perpendicular to an optic axis will therefore behave isotropically (black under crossed polars).

b) **INTERFERENCE COLOURS.** Each grain of a given mineral has its own specific interference colour (depending on the mineral species and the orientation of the crystal section, and assuming a constant thickness), but each mineral (when the component grains are viewed collectively) exhibits a range of colours. These colours form an ascending series (known as **Newton's Scale),** ranging from greys, whites and pale yellows, through repeated sequences of bright colours, and then into rather faded colours. For a given mineral it is the highest colour in this series that is diagnostic **(maximum interference colour)** but most grains of the same mineral will exhibit colours lower in the sequence. It is convenient to recognise three main categories of interference colours:-

i) **LOW** - greys, white and pale yellows
ii) **INTERMEDIATE** - bright reds, blues and greens
iii) **HIGH** - pale greys and browns with pink or green tinges.

(These interference colours should not be confused with the natural colour described in section 1b. In particular, it should be noted that interference colours will be modified in the case of strongly coloured minerals).

Investigate the following properties under **plane polars** (**PP**), using the slides indicated:

Relief

1. **RHB** contains roughly equal amounts of a **high-relief** mineral and a **low-relief** mineral. If you cannot see these straight away, then see what happens when you close the substage iris diaphragm. When you do this, the high-relief mineral will stand out from the low-relief mineral. As well as having dark outlines, you will see that the **high-relief** mineral contains prominent fractures that also stand out. You might also see some weak cleavage that stands out as fine dark parallel lines. The **low-relief** mineral may be fractured and have cleavage, but you cannot see this because of its low relief. Draw a sketch of a few of the high-relief grains, showing their fractures.

Colour

2. **EG** contains a few different minerals. One of them is **opaque**: it is completely black. As well as this, there is a **high-relief** mineral with fractures, and a **low-relief** mineral that may have some fractures, but they are hard to see because of the low relief (see 1 above for more information about relief).

3. **15216** is mostly made of low-relief, colourless minerals, but there is also a **brown, pleochroic** mineral (altered in places to a green mineral). Look for a crystal that is roughly rectangular in shape, and observe how the intensity of the colour changes as you rotate the stage (occasionally you will find a crystal that stays the same colour as you rotate the stage – it is unlikely to be rectangular). Draw 2 sketches of the crystal, one showing its darkest brown, one its lightest brown.

4. **NH** contains abundant crystals of an elongate **pleochroic** green mineral. Sketch an elongate crystal with its long axis oriented E-W and describe its colour. Now rotate the stage 90°, draw it again, and describe its colour. When describing the colour, as well as giving the colour (e.g. blue-green) you should say whether it is dark, light or something in between.

Cleavage

5. **15216**. Look again at the pleochroic brown mineral. You should be able to see a set of fine black lines parallel to the edge of the crystals. This is **cleavage** – planes of weakness in the crystal that cause it to split into parallel sheets. Add some cleavage lines to the sketches you drew in 3.

6. **VT.** There are two high-relief minerals in this sample. One is strongly pleochroic in shades of brown; the other has a much paler colour (with rims of green caused by the outer part of the crystal having a different composition from the rest, a property called **zoning**). Both show **cleavage**. The cleavage is quite variable in the paler mineral – some crystals do not show any cleavage, and most also have **fractures**. The cleavage is more obvious in the brown mineral. Look for a crystal that is roughly diamond-shaped. You should see two sets of cleavage planes in it. Sketch this crystal. Crystals cut in different orientations will only show one cleavage (but may also show fractures – make sure you can tell these apart from cleavage).

Investigate the following properties under **crossed polars** (**CP**), using the slides indicated:

Isotropy, anisotropy and interference colours

7. **CFL.** The mineral that is zoned in shades of purple (and much of it is colourless) is **isotropic**: it is black under cross polars and stays black as the stage is rotated. It is **not** opaque – it is transparent in PP light. The other mineral in this sample (that is colourless) has a **high interference colour**. Overall it is a pale sandy colour, but when you look closely you can see pink and green tinges. Notice how crystals go black as you rotate the stage – this is described below (8ii).

8. **E302.** This rock is made almost entirely of an **anisotropic** mineral with a **low interference colour**. Note the following: (i) Different grains show different shades of grey (ranging from almost black to white). This range of colours is due to different orientations of the crystal structure in each grain. (ii) Every grain goes into **extinction** (black) as you rotate the stage – this happens at 90° intervals. This property applies to all anisotropic crystals. (iii) Occasionally you will see a grain that appears isotropic – it stays black all the time as you rotate the stage. This happens because the crystal has been cut perpendicular to its optic axis (see property 2a above).

9. **R56.** The crystals in this thin section show a range of interference colours. Some are low, but most are **intermediate**: bright reds, blues and greens. Most of the crystals are of the same mineral, but because their crystal structures are in different orientations, they show different colours. Compare the colours you can see with the colours on the **Michel-Levy colour chart** (which shows Newton's scale of interference colours), to determine what is the highest interference colour for this mineral. With the analyser **out**, draw a sketch of a few grains of this mineral, noting its relief and whether it has cleavage or fractures. As with the **anisotropic** minerals in CFL and E302 (and all anisotropic minerals), the crystals in R56 go into extinction every 90° as you rotate the stage (with the analyser in), and occasionally you might see a grain that appears isotropic.
   R56 also contains a mineral with **low interference colour** that has a zebra-striped appearance. This is called **twinning**.

10. **R25 or SK** contains two minerals with contrasting interference colour: one is **low** and the other is **high.** Draw a sketch of a small area, annotating it to show which is which. The high interference colour mineral is the same as the one in CFL, and the low interference colour mineral is the same as the one in E302, so they have the same optical properties as the minerals in those thin sections.

Bringing everything together:

11. If you have time, go back to thin section **EG** (see 2 above), and describe all the optical properties of the high-relief and low-relief minerals (you cannot do any more with the opaque mineral).

12. How many different minerals can you see in **PC37**? Describe their optical properties as best as you can.

2-D sections through 3-D crystals / grains

